

Erika and Tim wrote the expression $\frac{(-2x^{-4} \cdot x^2)^3}{4x^{-5}}$ in different ways and arrived at different expressions. Their work is shown.

Erika's Work	Tim's Work
$\frac{(-2x^{-4} \cdot x^2)^3}{4x^{-5}}$ $\frac{-8x^{-12} \cdot x^6}{4x^{-5}}$ $\frac{-8x^{-72}}{4x^{-5}}$ $\frac{-2}{x^{67}}$	$\frac{(-2x^{-4} \cdot x^2)^3}{4x^{-5}}$ $\frac{(-2x^{-2})^3}{4x^{-5}}$ $\frac{-8x^{-6}}{4x^{-5}}$ $\frac{-2}{x}$

1. Even though one student's resulting expression is not equivalent to $\frac{(-2x^{-4} \cdot x^2)^3}{4x^{-5}}$, identify which laws of exponents each student applied in what order.

Order of Laws Applied	Erika	Tim
First	$(a^m)^n = a^{m \cdot n}$ Power of a Power	$a^m \cdot a^n = a^{m+n}$ Product of Powers
Second	$a^m \cdot a^n = a^{m+n}$ Product of Powers	$(a^m)^n = a^{m \cdot n}$ Power of a Power
Third	$\frac{a^m}{a^n} = a^{m-n}$ Quotient of Powers	$\frac{a^m}{a^n} = a^{m-n}$ Quotient of Powers

2. Which student made an error?

Erika

3. Write a note to this student explaining their mistake and how to correct their error.

When $a^m \cdot a^n = a^{m+n}$, Product of Powers, was applied on step 2, the exponents were multiplied instead of added. $x^{-12} \cdot x^6 = x^{-12+6} = x^{-6}$

For each expression in questions 4–10, write an equivalent expression using positive exponents and the fewest factors as possible.

$$4. \quad \frac{8w^{-6}p}{-2w^{-4}p^{-5}} \quad \frac{-4p^6}{w^2}$$

$$5. \quad \frac{k^{-1}j^{-6}}{7k^3j^{-2}} \quad \frac{1}{7k^4j^4}$$

$$6. \quad \left(\frac{-5f^{-6}g^3}{2f^7g^{-3}}\right)^2 \quad \frac{25g^{18}}{4f^{26}}$$

$$7. \quad (m^0 \cdot m^{-9})^{-4} \quad m^{36}$$

$$8. \quad c^{-5} \cdot \frac{3c^{-9}}{-6c^5} \quad -\frac{1}{2c^{19}}$$

$$9. \quad \left(\frac{3m^8n^{-9}}{-8m^{-11}n^2}\right)^0 \quad 1$$

$$10. \quad \frac{-7d^{-5} \cdot d^4}{11d^{-19}} \quad \frac{-7d^{18}}{11}$$